

EE P 564: TinyML

Lecture 9: Responsible AI in TinyML and Summary of Topics Covered

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**ELECTRICAL & COMPUTER
ENGINEERING**

UNIVERSITY of WASHINGTON



Topics Covered

- Introduction to Responsible AI
- Responsible AI in TinyML
- Privacy and Responsible AI
- Summary of Topics Covered
- Applications of TinyML
- The Future of TinyML



Problems of AI

AI offers great benefits, but at the same time it introduces new risks and ethical challenges

Artificial intelligence can...

- Change current practices
- Influence human decisions
- Regulate human behavior



Problems of AI - Examples

Microsoft's disastrous Tay experiment shows the hidden dangers of AI

Google Calls Hidden Microphone in Its Nest Home Security Devices an 'Error'

Predictive policing algorithms are racist. They need to be dismantled.

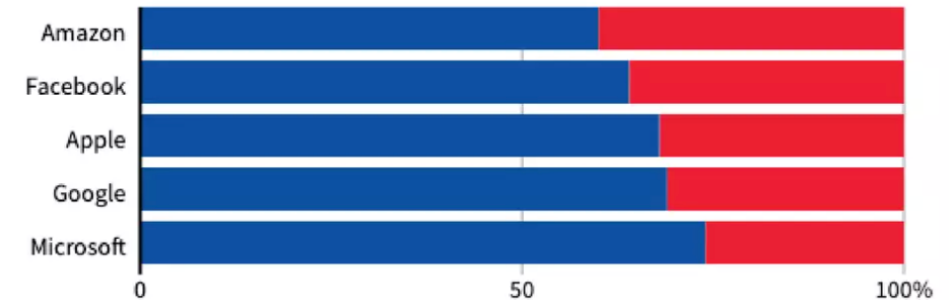
Problems of AI - Examples

Recruiting Tools

Amazon recruiting tool shut down for bias against women after it codified discriminatory practices due to narrow data sets.

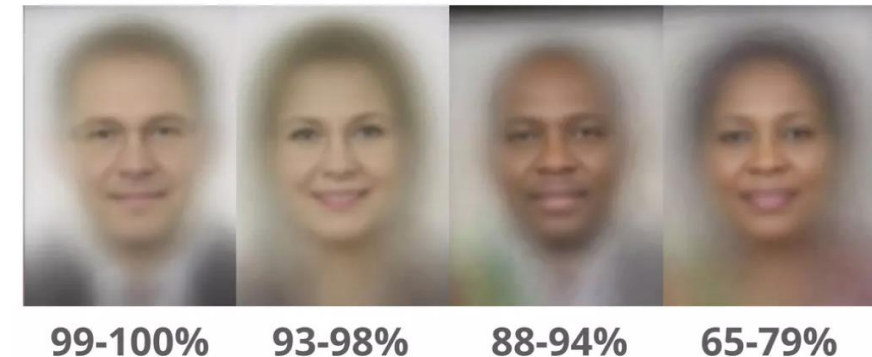
GLOBAL HEADCOUNT

■ Male ■ Female



Recognition AI

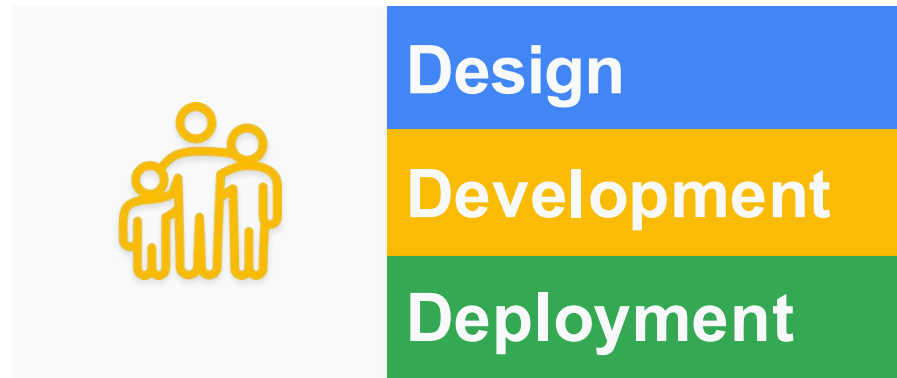
Calls for regulation on use of facial recognition after consistently higher error rates for darker-skinned and female faces.



What can we do?

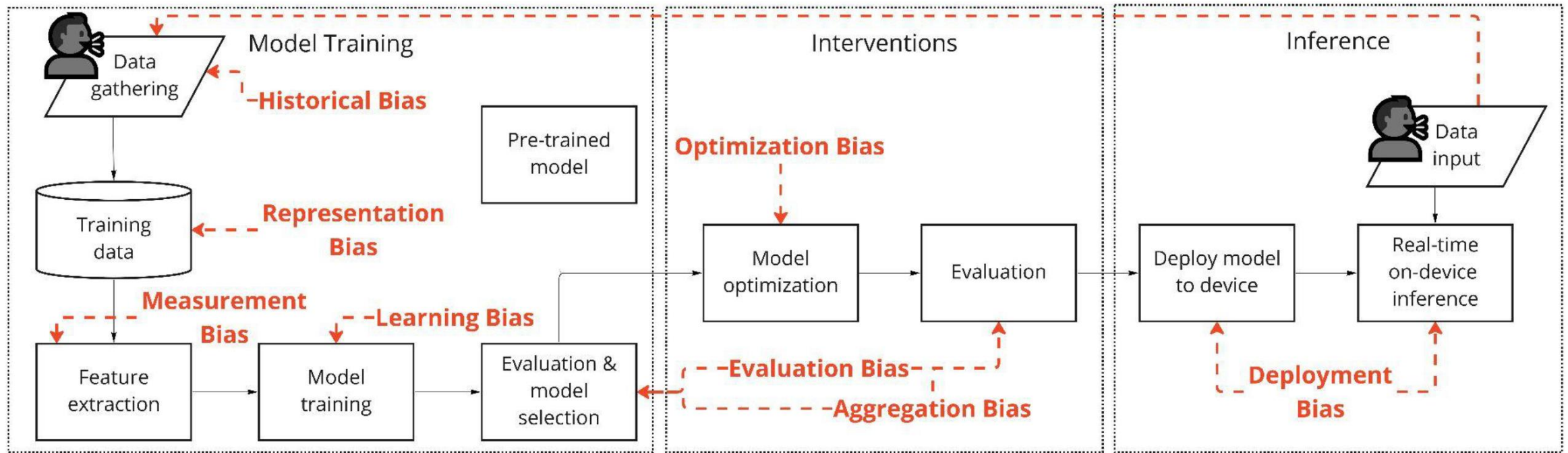
Responsible AI

- As creators of artificial intelligence systems, we have a duty to guide the development and application of AI in ways that fit our social values
- There is a growing trend towards creating **"human-centered" AI**
- Ethics is no longer just **reactive** but **proactive**

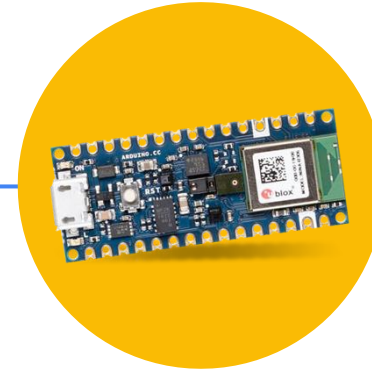


- This means keeping human values in the loop throughout all stages of a product's lifecycle

Bias in TinyML



Responsible AI – Human Centering Design



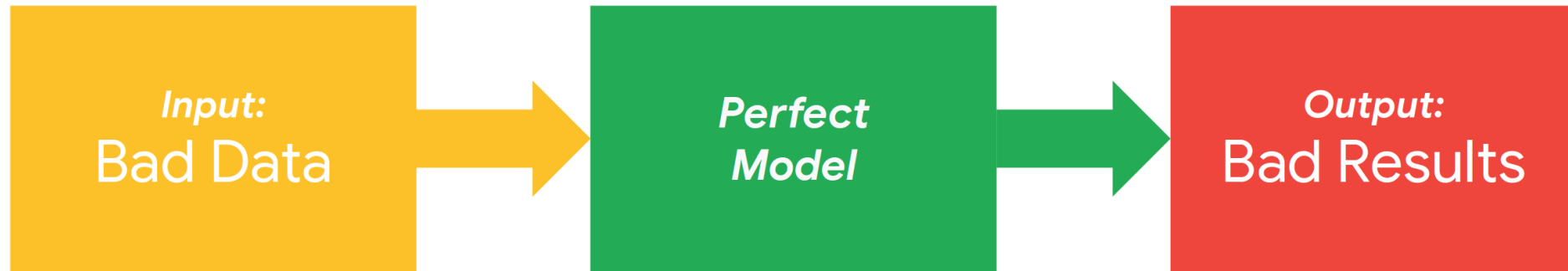
Responsible AI – Human Centering Design



- **What** am I building?
- **Who** am I building this for?
- What are the **consequences** for the user if it **fails**?
- **What data** will be collected to train the model?
- Is the dataset **biased**?
- How can we **ensure** the model is **fair**?
- How will **model drift** be monitored?
- How should **security breaches** be addressed?
- How should the user's **privacy** be protected?

Biased Dataset

Garbage in, Garbage out!



Example of Biased Dataset

1. Define the target variable

Using biometric sensors for a health wearable device, how should you define “healthy”?

- Heart Rate
- Blood Pressure
- Number of Steps



Example of Biased Dataset

2. Label the data



Horse



Human



Human



Labels applied to the training data must serve as ground truth !

Example of Biased Dataset

3. Prejudice Reflected in Data



Dataset: 65% of people cooking are women

Algorithm predicts: 85% of people cooking are women

Solutions from Industry

- Google Research Initiative to collect data and refine speech recognition algorithms to work better for individuals with speech impairments



Solutions from Industry

- Common Voice Dataset
- **Accent:**
23% United States English, 8% England English, 5% India and South Asia, 4% Australian English, 3% Canadian English, 2% Scottish English, 1% Irish English, 1% Southern African, 1% New Zealand English
- **Age**
23% 19–29, 14% 30–39, 10% 40–49, 6% < 19, 4% 50–59, 4% 60–69, 1% 70–79



Privacy and Responsible AI

Fit Leaking

Citizen Lab Research on Fitness Tracker Privacy



Image showing pattern-of-life activity in a suspected Russian military installation on an airbase in Syria. Image: Strava.



Human Visual Processing: Limited Bandwidth, Inherent Bias

Stage	Approx. Data Rate	Description
Retinal Input	~10 million bits/sec	Raw visual input captured by rods and cones in the retina
Neural Processing	~1 million bits/sec	Filtered, pattern-recognized input handled unconsciously
Conscious Awareness	~40–60 bits/sec	What we actually perceive, attend to, and reason about

 **Because our conscious brain processes only a tiny slice of incoming information, human perception is inherently selective — and with that selectiveness comes bias.**

Why is privacy valuable?

- **Prevent information-based harms**
 - Minimize opportunities for hackers to gain inappropriate access to data
- **Prevent informational injustice and discrimination**
 - Consider the context, the type of information, and who has access
- **Preserve autonomy and human dignity**
 - Obtain informed consent

How can privacy be preserved?

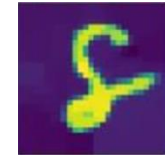
- **Minimize**
 - Avoid collecting unnecessary data, and dispose or delete data periodically
- **Protect**
 - Use encryption techniques to protect data
- **Map the flow of Information**
 - Context, the type of information, and who has access
- **Information Consent**
 - Be transparent with users about how their data is being collected and used

Privacy Attacks on Deep Learning

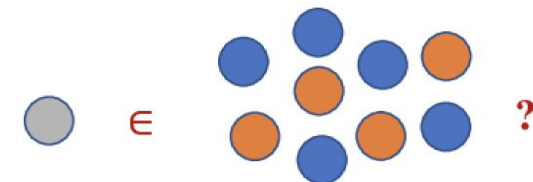
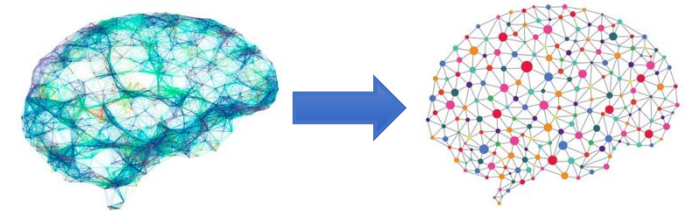
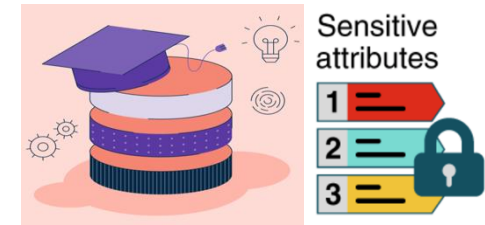
- Recreate one or more training samples
- Extract dataset properties which were not explicitly encoded as features
- Create a substitute model that learns the same task
- Check whether an input sample was used as part of the training set



Original



Reconstructed

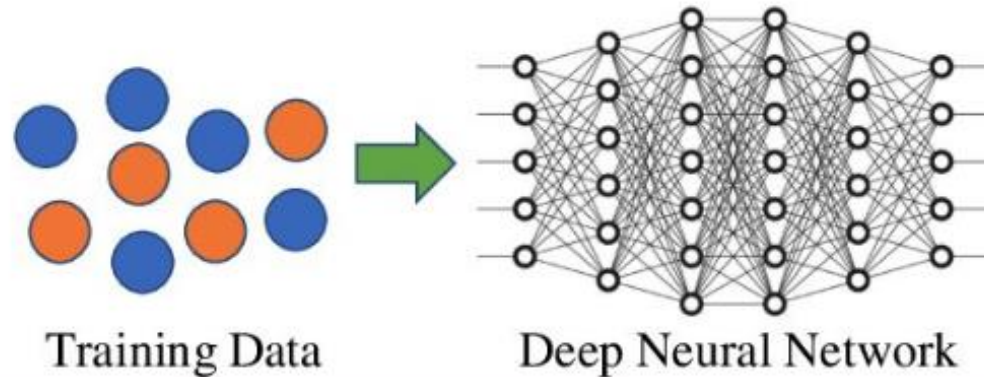


Training Data

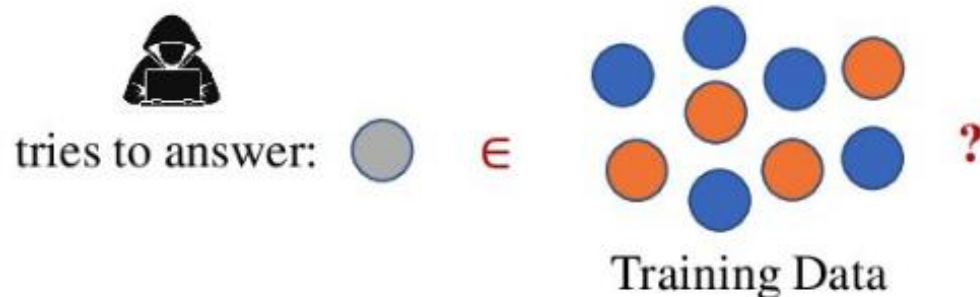
Membership Inference Attack (MIA)

- A widely studied privacy attack in machine learning
- Allow an adversary to **determine whether a specific data point was part of the training set**

Training of Target Model



Membership Inference Attack on Target Model



- **Member:**
Input sample **belongs to training set**
- **Nonmember:**
Input sample **does not belong to training set**

Examples of MIAs

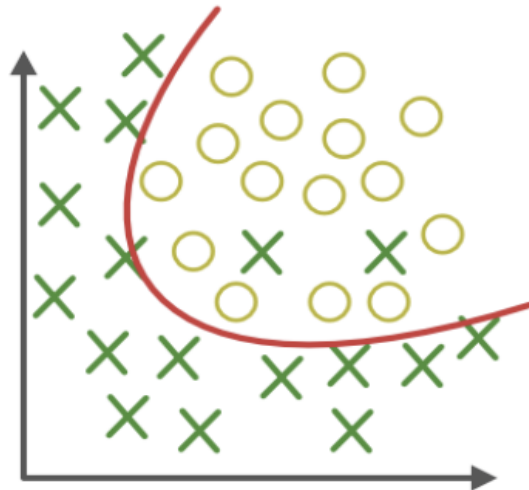
- Identifying the **hospital** where **medical inputs** (e.g., X-rays, ECGs) were taken
- Recognizing the **citizenship** of a **person** in an image
- Finding out the name of the **company** that input **financial data** belongs to



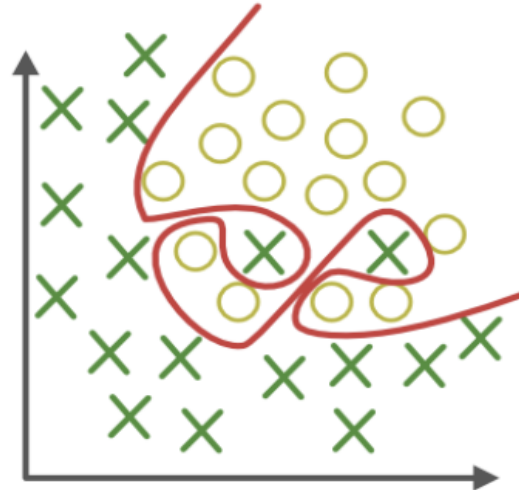
Insight of MIAs

Overfitting is the reason for the feasibility of mounting MIAs

- DNN models **perform extremely well on member** samples by **memorizing** patterns in the data → **Higher confidence** scores
- DNN models do not demonstrate the same performance on **nonmember** samples → **Lower confidence** scores



Appropriate fitting



Overfitting

Confidence-based MIAs

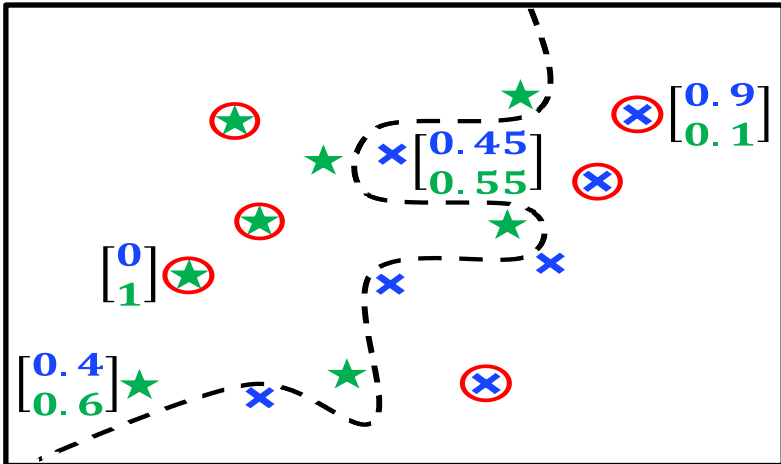
- Confidence scores are used by adversary to distinguish between members and nonmembers (e.g., GAP attack)

- Attack Success Rate (ASR) of GAP attack:

$$ASR_{GAP} = 0.5 + \frac{Acc_{mem} - Acc_{nonmem}}{2}$$

- Acc_{mem} : Classification accuracy of members
- Acc_{nonmem} : Classification accuracy of nonmembers

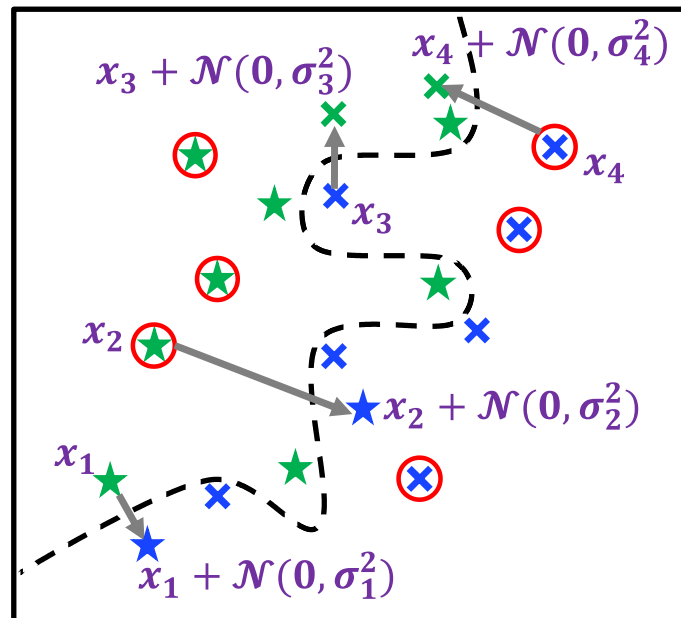
- Confidence score masking defenses (e.g., MemGuard) hide true confidence values to thwart confidence-based MIAs, but they are vulnerable to label-based MIAs



- **Member** samples are circled in **red**
- $\begin{bmatrix} \# \\ \# \end{bmatrix}$: Confidence vector

Label-based MIAs

- Adversary **perturbs** a data sample with **additive noise** until the newly generated sample is **misclassified**
- It then **estimates** the distance of the sample to the **decision boundary** based on the magnitude of the added noise
- Data samples **farther away from the boundary** (high confidence samples) are identified as the **members**



- **Member** samples are circled in **red**
- $\mathcal{N}(0, \sigma^2)$: Zero-mean Gaussian noise samples with variance σ^2
- $\sigma_1^2, \sigma_3^2 < \sigma_2^2, \sigma_4^2$

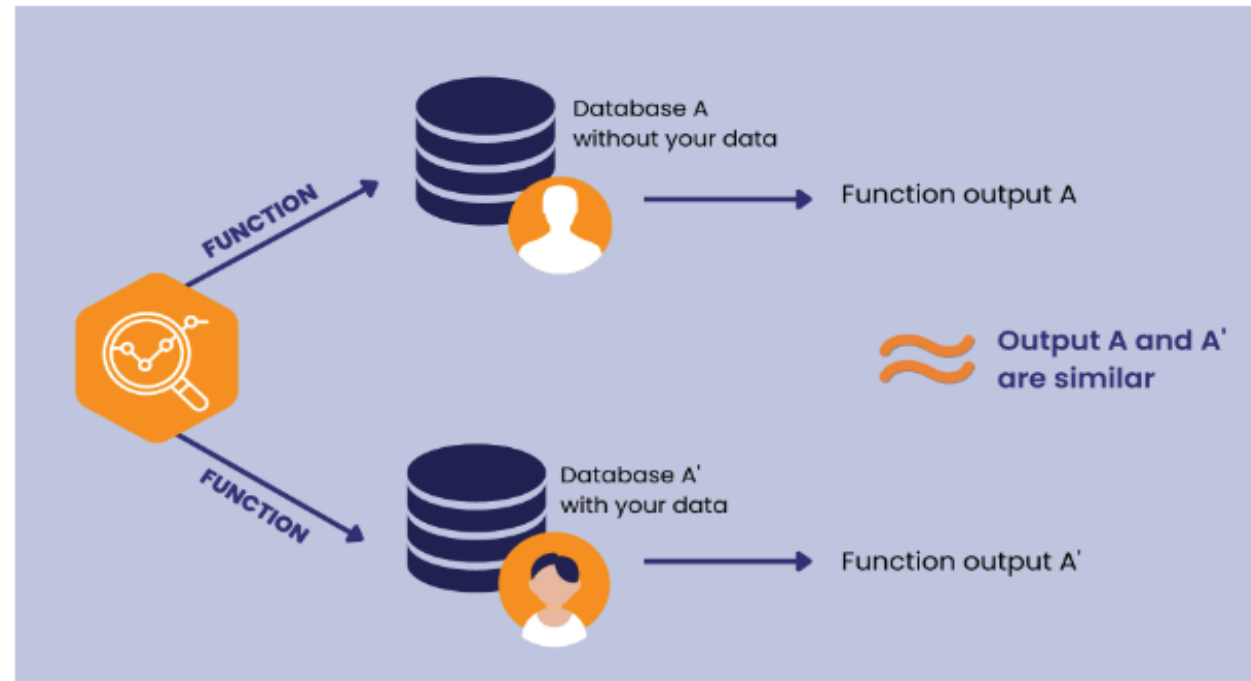
Defenses Against LAB MIAs

- **Differential Privacy (DP)** [Chen et al., 2020], [Truex et al., 2019]:
perturb model weights during training
- **L1 and L2 Regularization** [Ying et al., 2020]:
limits the values of the model weights during the training
- **Dropout** [Salem et al., 2019]:
disregarding certain nodes in a layer at random during training



Introduction to Differential Privacy

- Aims to provide **insights** about the dataset without revealing information about any individual data point
- ML algorithm is considered differentially private **if the probability of any outcome occurring is nearly the same for any two datasets that differ in only one record**



Formal Definition of Differential Privacy

- An algorithm A is (ϵ, δ) -differentially private if for all datasets D_1 and D_2 differing by one element, and for all possible outputs S of the algorithm:

$$\Pr[A(D_1) = S] \leq e^\epsilon \cdot \Pr[A(D_2) = S] + \delta$$

- Parameters:
 - ϵ : Privacy loss parameter or privacy budget. Lower ϵ means stronger privacy
 - δ : Probability of the algorithm not being ϵ -differentially private



Introduction to DP-SGD Algorithm

- Stochastic Gradient Descent (SGD):
 - A popular optimization method for training machine learning models
 - Iteratively updates model parameters to minimize the loss function
- DP-SGD: A variant of SGD that ensures differential privacy by incorporating noise and clipping gradients



Mechanism of DP-SGD Algorithm

- **Noise Addition:** Adds Gaussian noise to gradients during each update to obscure individual data contributions
- **Gradient Clipping:** Limits the influence of any single data point by clipping gradients to a maximum norm C
- **Algorithm Steps:**
 - Initialize model parameters (e.g., model weights)
 - For each batch
 1. Compute per-sample gradients
 2. Clip the gradient by norm C
 3. Add Gaussian noise
 4. Update model parameters with the noisy gradients



Benefits of DP-SGD Algorithm

- **Privacy Protection:** Safeguards against membership inference attacks by adding noise to the training process.
- **Scalability:** Applicable to large datasets and deep learning models.
- **Compatibility:** Integrates with existing ML frameworks (e.g., TensorFlow Privacy).



Challenges and Trade-Offs of DP-SGD Algorithm

- **Privacy vs. Utility:** Higher privacy (lower ϵ) often leads to reduced model accuracy
- **Hyperparameter Tuning:** Selecting appropriate values for noise scale and clipping norm is complex and critical for balancing privacy and utility
- **Computational Overhead:** Adding noise and clipping gradients can increase computational requirements



Real-World Applications of DP-SGD Algorithm

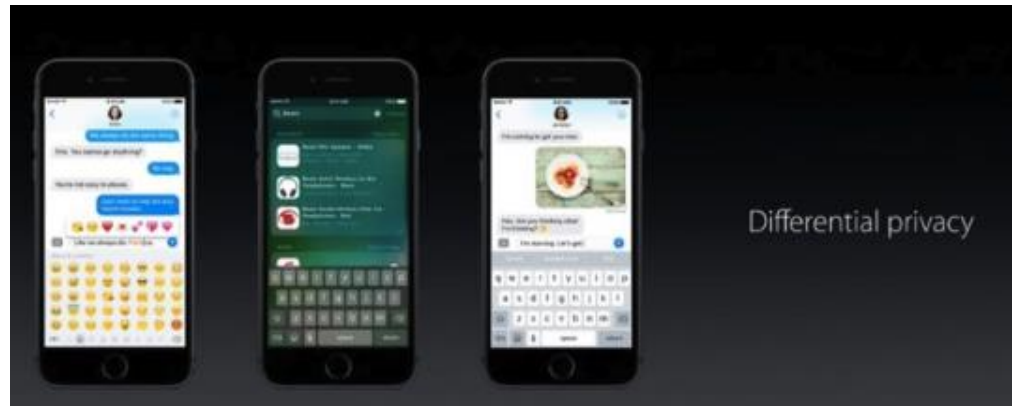
- **Google's RAPPOR:** Uses differential privacy to collect aggregate data from users while preserving individual privacy

google/**rappor**

RAPPOR: Privacy-Preserving Reporting Algorithms



- **Apple's Differential Privacy Implementation:** Utilizes differential privacy to collect data for improving user experience without compromising individual privacy



Defenses Against LAB MIAs Recap

- **Differential Privacy (DP)** [Chen et al., 2020], [Truex et al., 2019]:
perturb model weights during training
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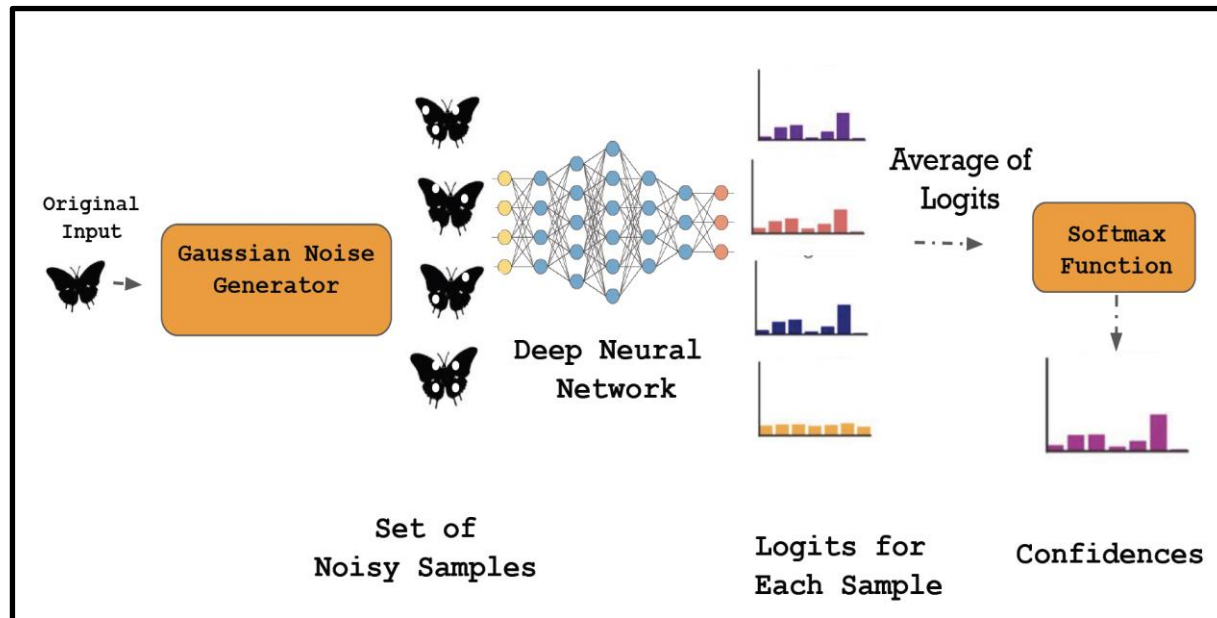
❖ However, these defenses require re-training the DNNs

❓ *Can we defend against LAB MIAs without having to re-train the DNNs?*



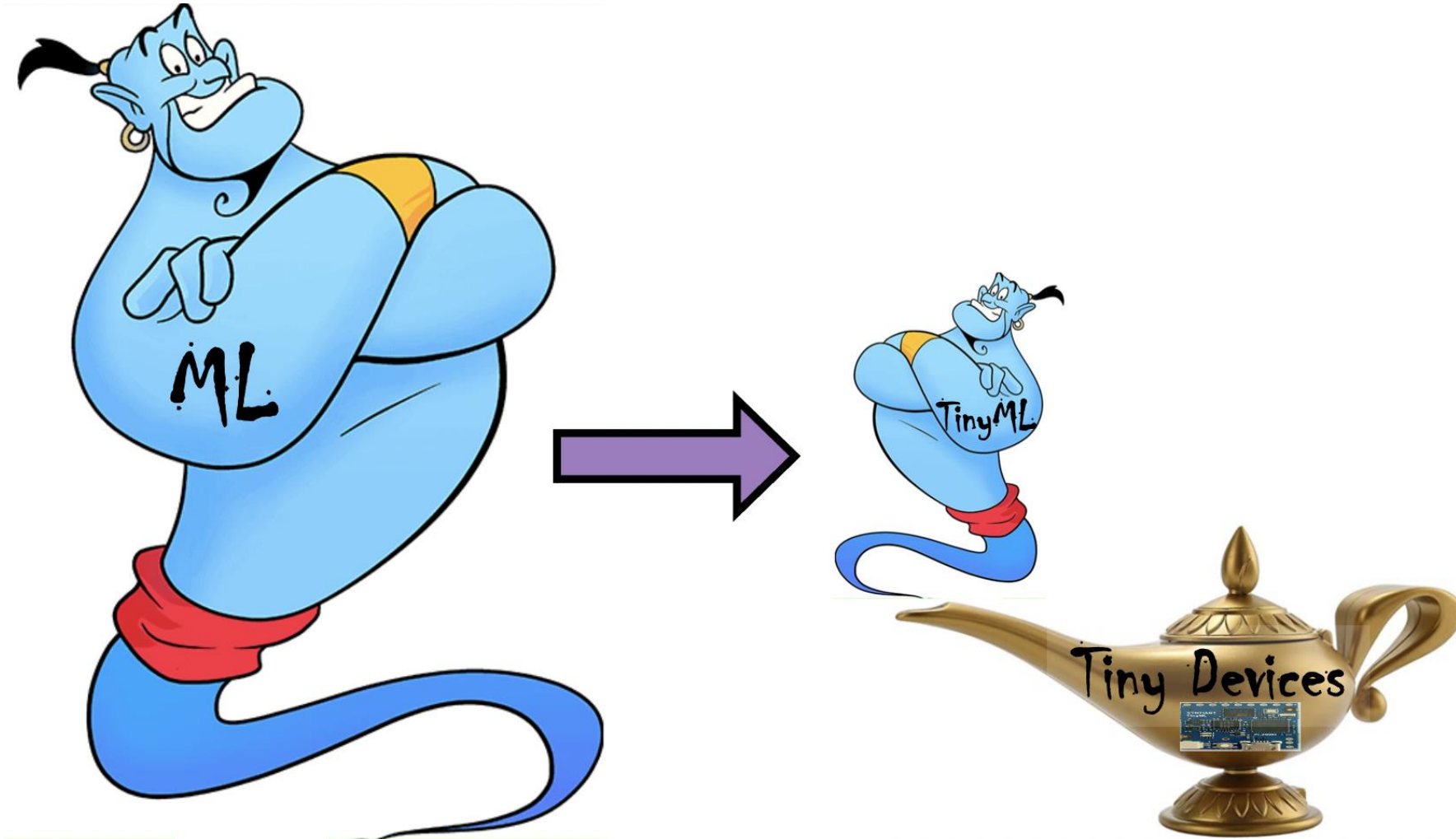
LDL: A Defense Against MIAs at Inference Time

- **Insight:** ensure that magnitudes of noise required for misclassification of members and nonmembers are comparable
- Uses randomization at inference time and constructs a high dimensional sphere of label invariant around the samples

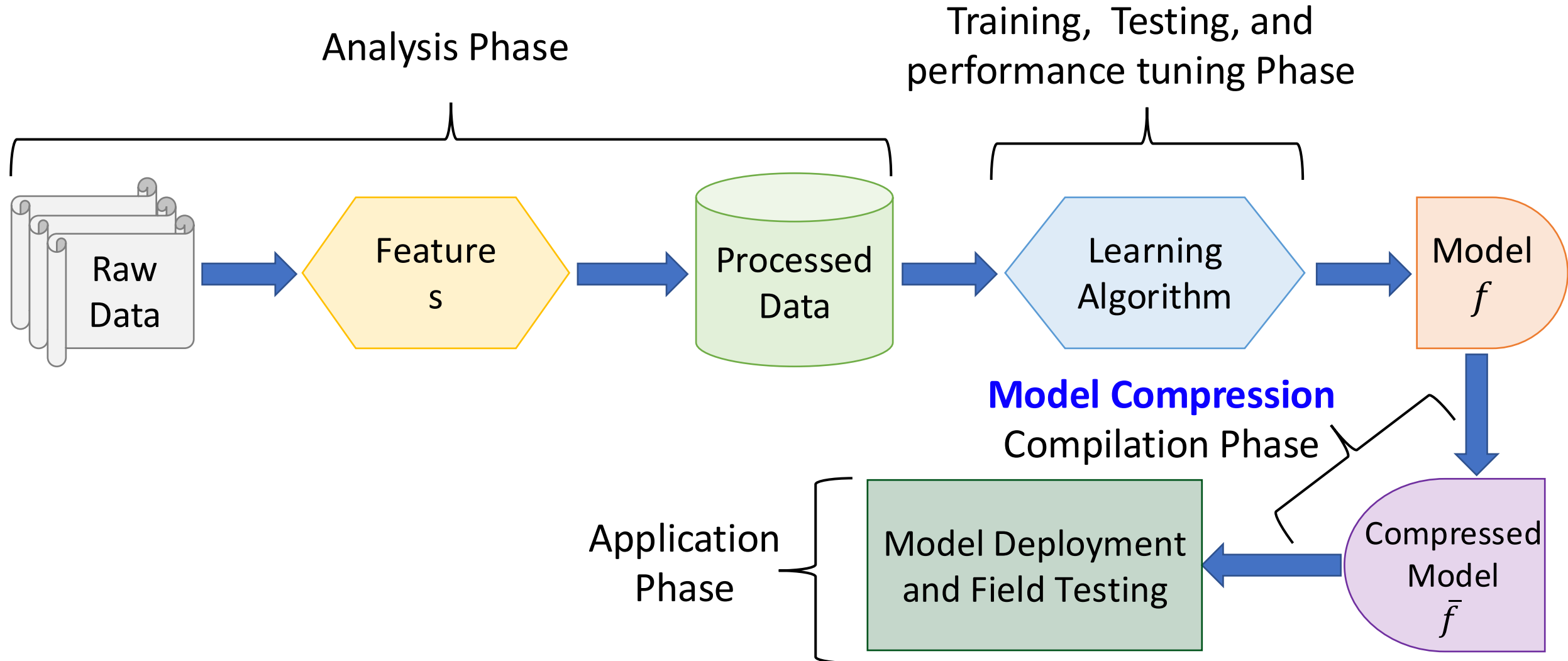


Rajabi, Arezoo, Dinuka Sahabandu, Luyao Niu, Bhaskar Ramasubramanian, and Radha Poovendran. "LDL: A Defense for Label-Based Membership Inference Attacks." In *Proceedings of the 2023 ACM Asia Conference on Computer and Communications Security*, pp. 95-108. 2023.

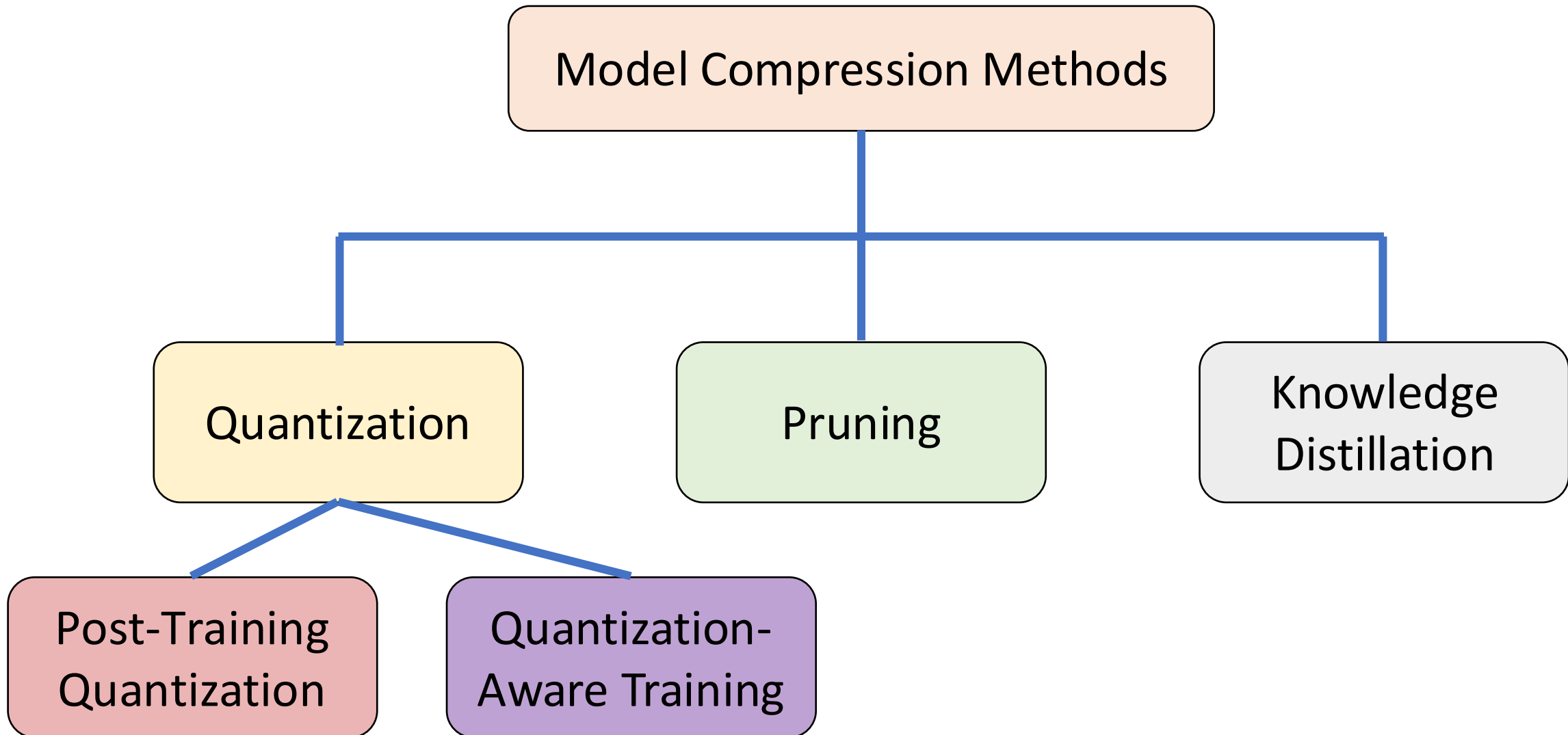
Review of TinyML



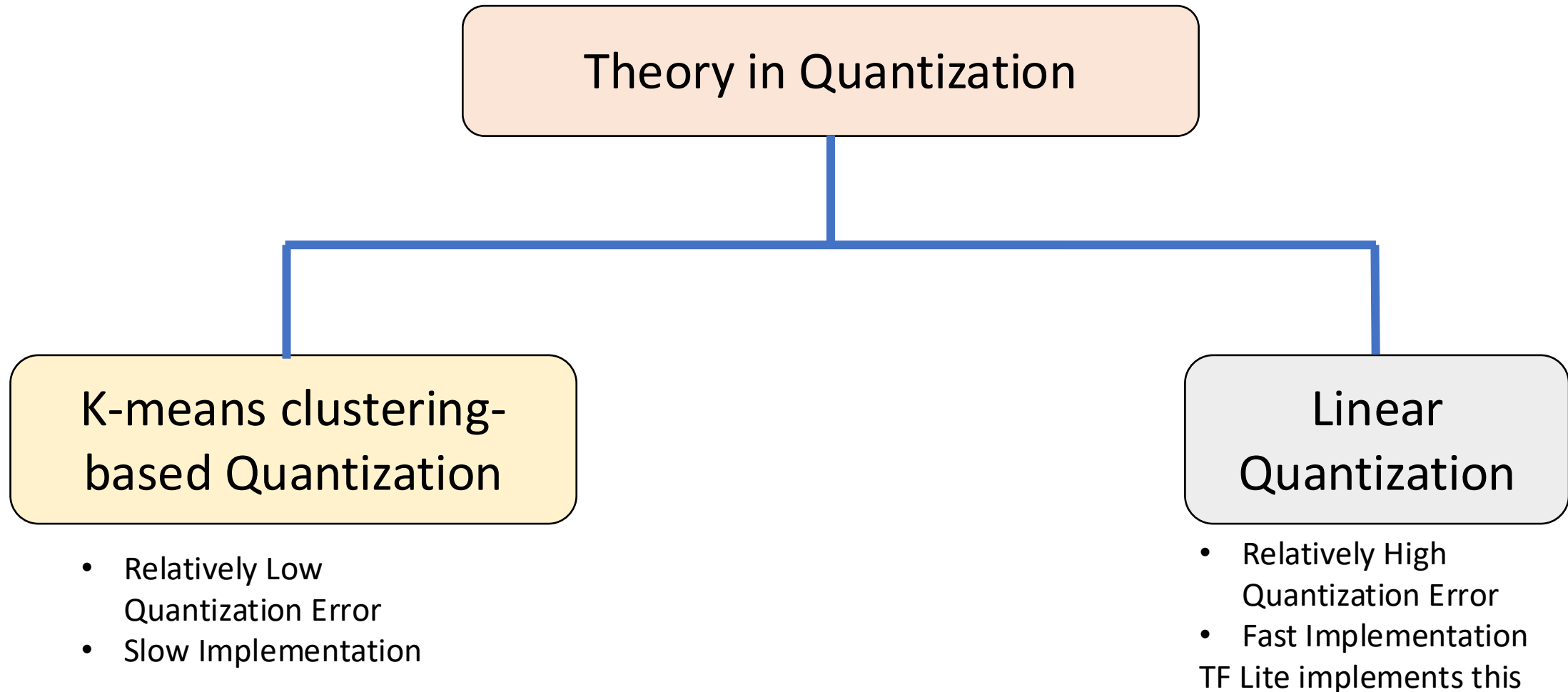
Review of TinyML



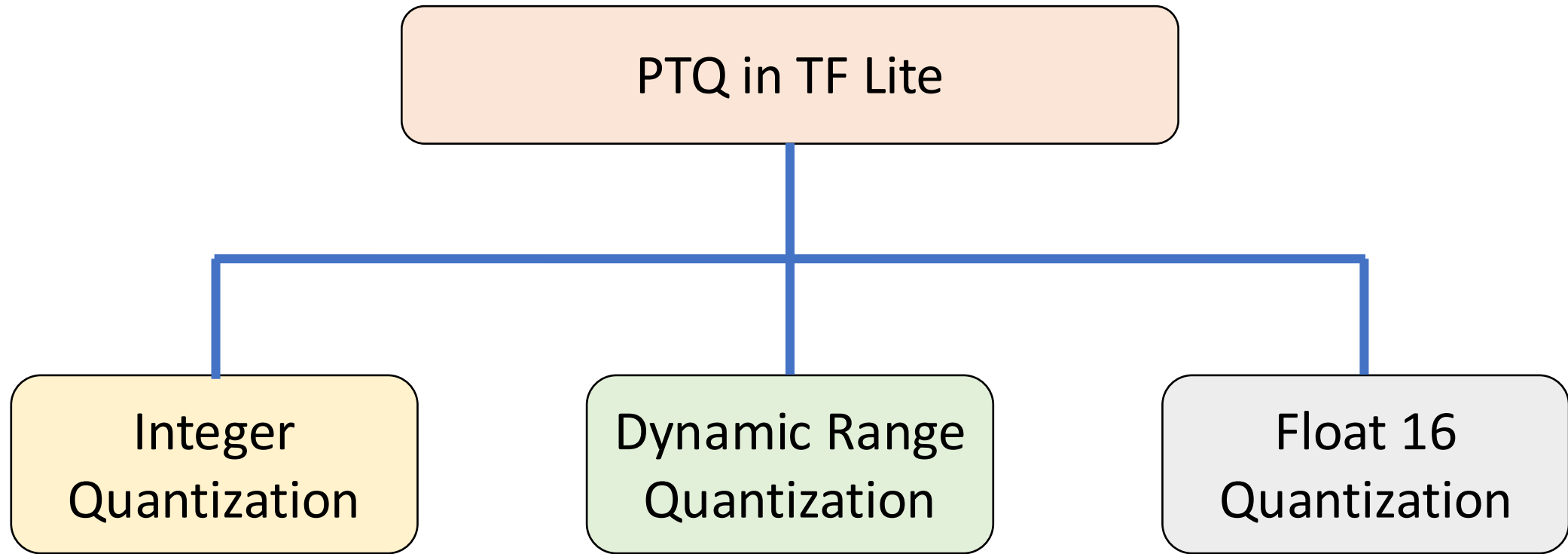
Review of TinyML – Model Compression Methods



Review of TinyML – Theory in Quantization



Review of TinyML – Post Training Quantization (PTQ)



Review of TinyML Techniques Learned in this Class

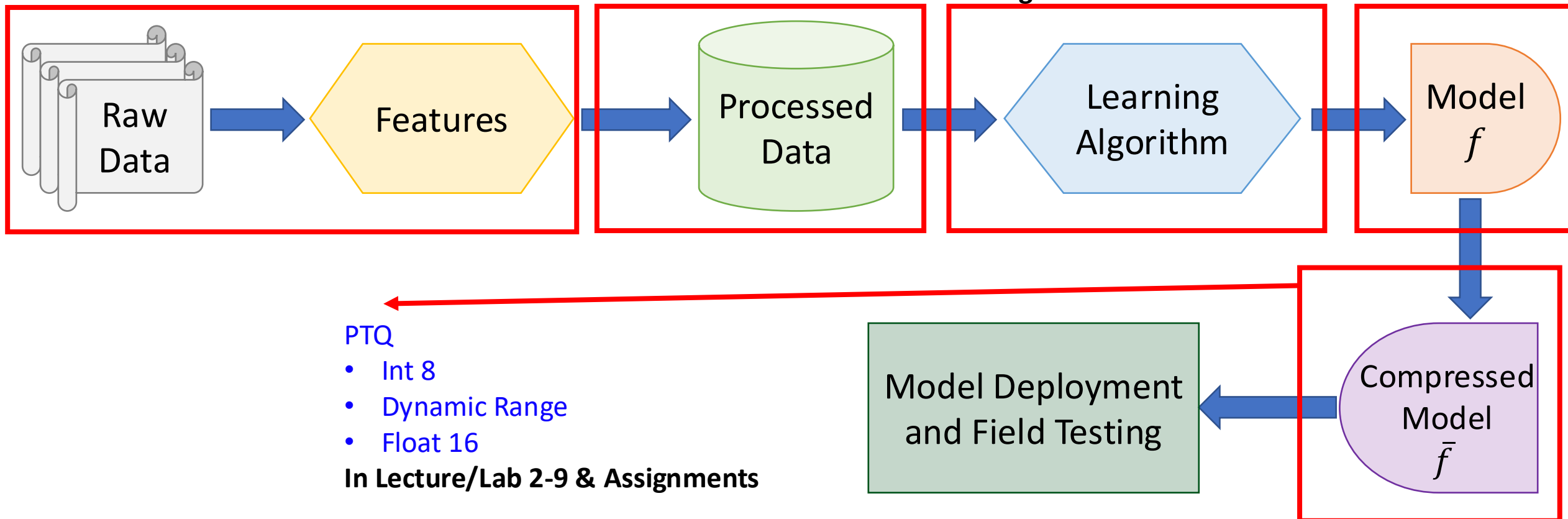
(If possible) **converting data into images** as CNNs support model compression techniques such as pruning across all the layers

In Lecture/Lab 3 and 6

- **Data Scaling** (min-max; standard) – **In Assignments**
- **Dimensionality Reduction** (PCA) – **In Lecture/Lab 5**

- **QAT** (int 8, dynamic)
 - **Pruning** (PCA)
 - **Knowledge Distillation**
- In Lecture/Lab 2 & Assignments**

Computation Efficient Neural Architecture Selection (mobile-net)
In Lecture/Lab 4



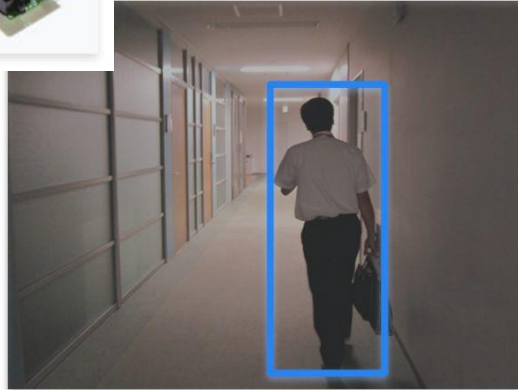
Review of TinyML - Applications

Sound



Keyword Spotting

Vision



Visual Wake Words
Mask Detection

IMU Sensors



Anomaly Detection
Magic Wand,
Motion Classification
American Sign Language

EEP 564 Course Summary

Main learning objectives

- ✓ Develop TinyML models to solve real-world problems
- ✓ Implement TinyML applications & explanation of needed ML algorithms
- ✓ Use TensorFlow for deep learning and TensorFlow Lite (TFLite) for TinyML
- ✓ Using C language for deploying TinyML on Embedded Systems
- ✓ Measure the performance of the deployed TinyML models

Each week we

1. Focused on a specific TinyML applications
2. Showed how to use TinyML to address the problem
3. Provided hands on experience in the labs

Course Summary: Topics Covered

- **Week 1: Introduction to TinyML**
 - TinyML Landscape, Applications and Challenges
 - TinyML Lifecycle and Workflow
 - Model Compression Techniques
 - Recap on Necessary ML Background: ML Algorithms, Neural Networks
 - Introduction to Hardware and Software Used in the Course
- **Week 2: Fundamentals of ML and TinyML**
 - Pruning ML models
 - Quantization Aware Training (QAT) and Post Training Quantization (PTQ)
 - Knowledge Distillation
 - Tiny Deep Learning
 - TensorFlow Lite (TFLite) for TinyML

Course Summary: Topics Covered

- **Week 3: TinyML for Keyword Spotting**
 - Background on Keyword Spotting and Streaming Audio
 - Challenges and Constraints in Keyword Spotting
 - Keyword Spotting Architecture and Data Collection
 - Model Training, Evaluation Metrics, and Deployment
- **Week 4: TinyML for Visual Wake Words**
 - Introduction to Visual Wake Words and Its Challenges
 - Visual Wake Words Dataset
 - MobileNets
 - Transfer Learning for Visual Wake Words
 - Model Training, Evaluation Metrics, and Deployment

Course Summary: Topics Covered

- **Week 5: TinyML for Anomaly Detection**
 - Background on Anomaly Detection and Signal Processing
 - Real and Synthetic Datasets
 - Unsupervised Learning (K-Means Clustering and Autoencoders)
 - Threshold Choice
 - Model Training, Evaluation Metrics, and Deployment
- **Week 6: Wizard Magic Wand**
 - Gesture Tracking through Bluetooth
 - CNN for Magic Wand Sketch
 - Data Collection and Labeling
 - Model Training, Evaluation Metrics, and Deployment

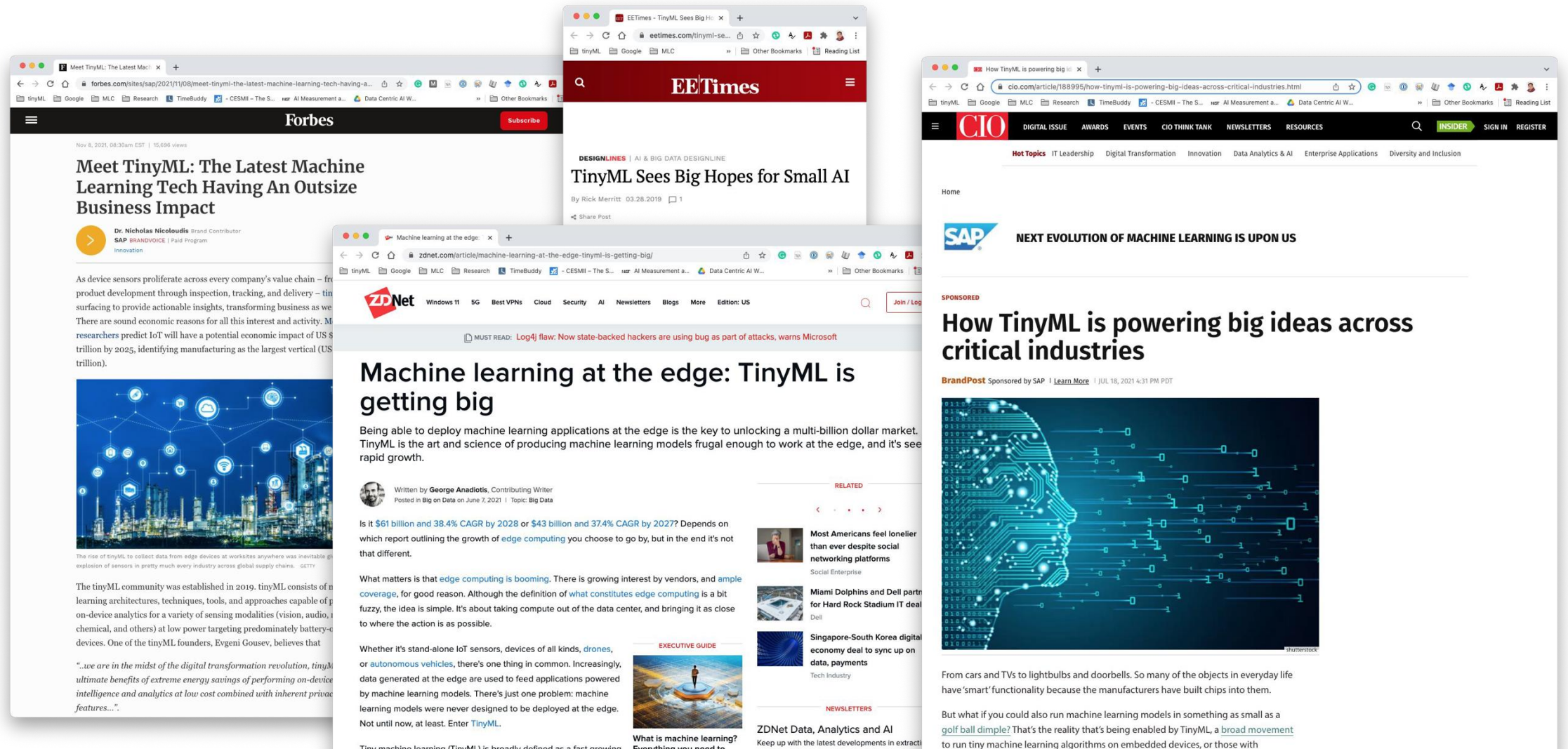
Course Summary: Topics Covered

- **Week 7: TinyML for Motion Classification**
 - Background on Motion Classification
 - Accelerometer, Gyroscope, Barometer, and Magnetometer
 - Supervised learning for motion classification
 - Sensor interface for Mechanical Stress in Transport
 - Model Training, Evaluation Metrics, and Deployment
- **Week 8: TinyML for American Sign Language (ASL) Interpretation**
 - Background on ASL and ASL Interpretation
 - Gesture Motion Datasets and Features
 - Analyzing Gesture Motion Data using Neural Networks
 - TinyML Framework for ASL Interpretation
 - Model Training, Evaluation Metrics, and Deployment

Course Summary: Topics Covered

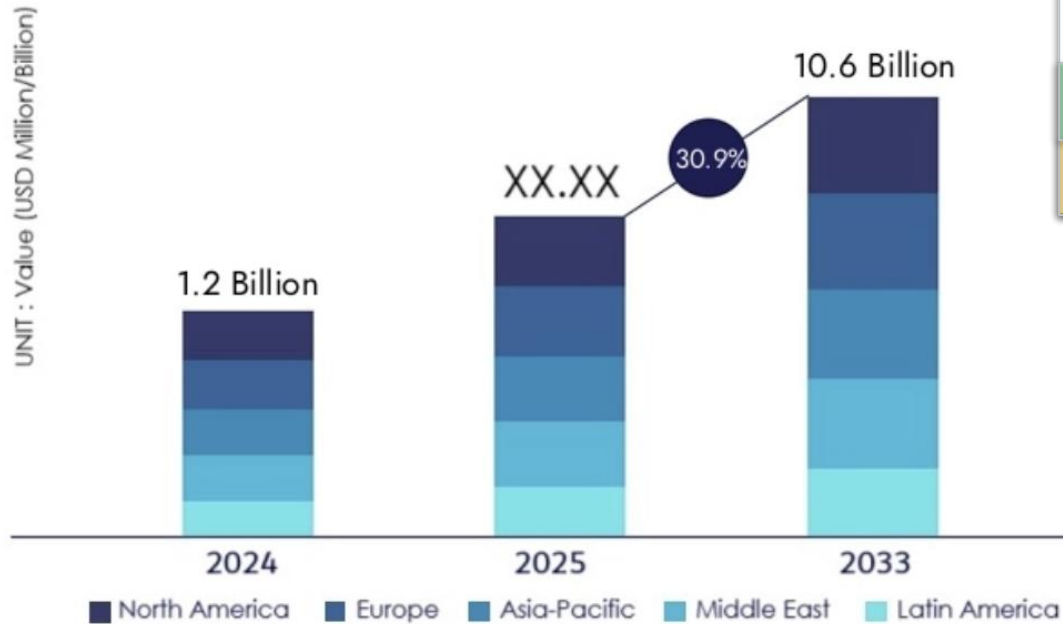
- **Week 9: Responsible AI in TinyML and**
 - Introduction to Responsible AI
 - Responsible AI in TinyML
 - Privacy and Responsible AI
 - Summary of Topics Covered
- **Week 10: No Lecture Meetings**
 - Use the regular class time to wrap up your final projects.
 - TAs will hold additional office hours during Week 10.
- **Week 11: Final Presentations and Demos**
 - **Date and time:** To be announced by early Week 10.

The future of the TinyML



The future of the TinyML

Global Tiny Machine Learning (TinyML) Market Size and Scope



As device sensors proliferate across every company's value chain – from new product development through inspection, tracking, and delivery – **tinyML** is surfacing to provide actionable insights, transforming business as we know it. There are sound economic reasons for all this interest and activity. **McKinsey researchers** predict IoT will have a potential economic impact of US \$4-11 trillion by 2025, identifying manufacturing as the largest vertical (US \$1.2-3.7 trillion).

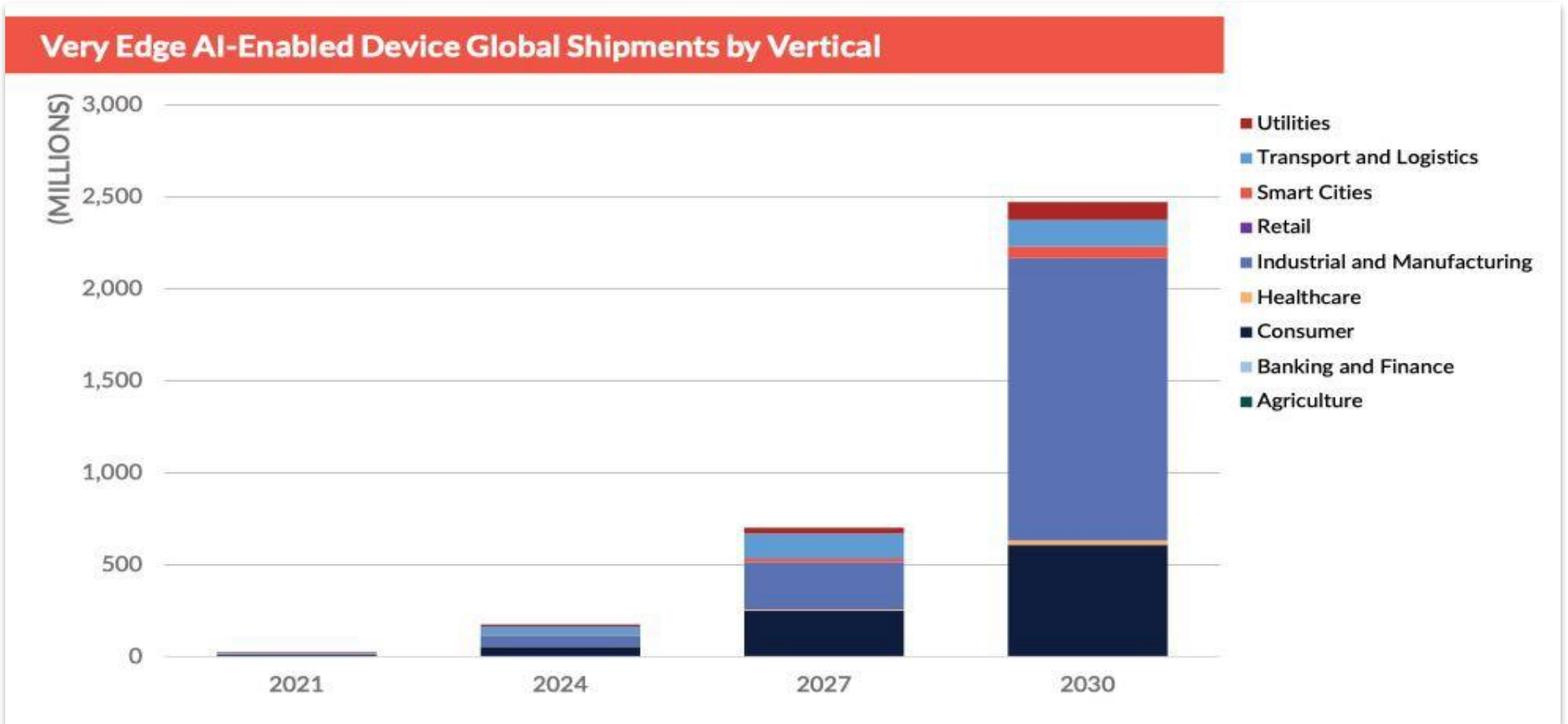
Source: Forbes

30.9%
CAGR from 2026 to 2033

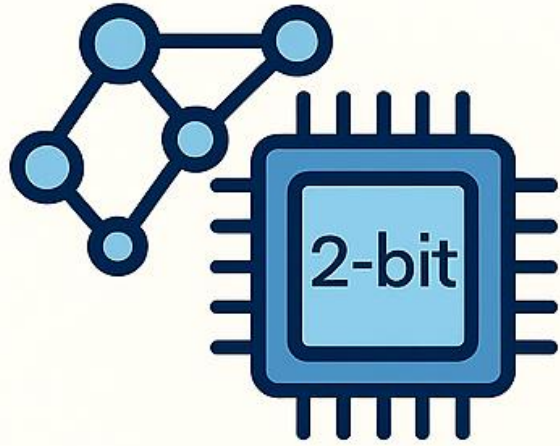
Global Tiny Machine Learning (TinyML) Market size was valued at 1.2 billion USD in 2024 and is projected to reach 10.6 billion USD by 2033

Source : www.verifiedmarketreports.com

The future of the TinyML



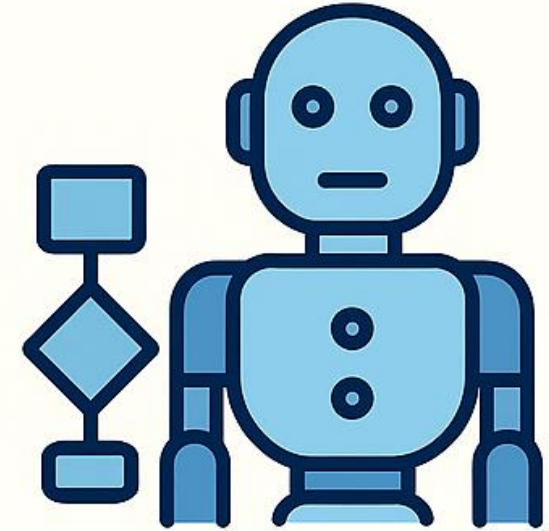
The Future of the TinyML



**Ultra-Low-Bit
LLMs on
FPGAs and
Microcontrollers**



**AI
Alignment**



**Agentic
AI**

Thank You!

The Future of ML is Tiny and Bright

A Quote from Prof. Vijay Janapa Reddi (Harvard University) and Pete Warden (Google)